

QuarkWave hardware test log

Purpose: record repeatable results for the assembled Pico W and Uno R4. The guides are **source-reviewed, hardware untested** until the relevant checks pass. Copy this page for each firmware and wiring revision; record actual observations rather than replacing expectations with a checkmark. The dated records below are chronological: older entries describe earlier builds and are superseded by later entries, not by edits to their measurements.

Test record

Field	Record
Date, tester, location	
Pico / Uno board and core versions	
Firmware revision or the three hashes in the index	
Arduino libraries and relevant versions	
Wiring revision / connection guide evidence	
Power sources; measured Pico RX voltage during Uno TX	
Audio destination and listening level	
DIN controller, RTP-MIDI host, browser/device used	

For each check, record **Pass / Fail / Not run**, the observed behavior, and a short note or evidence link. If a check fails, include the smallest repeatable steps and any MIDI, serial, or screen capture that helps reproduce it.

Check	Expected from source	Result / observation
Both boards start together	Uno replies when ready; Pico sends the selected boot patch on a clean Uno. Browser timing does not trigger sync.	
Uno starts late; Uno restarts	Pico detects the new ready state and syncs a clean Uno.	
Pico starts late; Pico restarts	Standalone Uno sound persists if an external input was used; otherwise the selected Pico patch syncs.	
Return MIDI wire disconnected	Uno connection label goes offline after missed replies; no false patch acknowledgment.	
Browser opens before / after sync	Pico and Uno status labels remain separate and consistent.	

Check	Expected from source	Result / observation
Browser note and release	Pointer and computer keys send notes; release and browser disconnect end owned notes.	
Keyboard Velocity 1 / 100 / 127	New notes use the selected velocity; held notes keep the old velocity.	
Perform / Shape / Explore / All controls / Help	Controls keep values; held browser notes continue across view changes. Help pages make no sound change.	
Factory Load and user Save As	Factory sounds stay read-only; Save As chooses a slot and confirms occupied replacement.	
User Save, Load, restart	Saved file, selected slot, and recovered sound agree; failure does not change selection.	
Commit / Load committed sound	Snapshot can be saved and restored separately from user slots.	
Randomize	Returned panel values remain inside each control's range, including effects.	
Panic and visualizations	Panic silences notes; Status, VU, and Scope select the intended Uno LED view.	
DIN without Pico	Notes, bend, sustain, and CC work with Uno defaults; no Pico is required.	
RTP-MIDI through Pico	The Pico's <code>QuarkWave</code> session accepts notes and sound controls, then forwards them to the Uno. Without the Pico, no RTP-MIDI session exists in this build.	
Standalone then attach Pico	Existing sound remains until explicit Sync Pico patch to Uno ; warning appears.	
External Clock and transport	40 / 120 / 240 BPM, Start / Continue / Stop, DIN priority, two-second failover, and held tempo behave as documented.	
External custom messages	Accepted sound Program Changes and SysEx change sound and mark standalone activity; Pico-only commands have no effect on DIN/RTP.	
Output and levels	Listen and measure across master volume, effects, and four-note/unison cases. Document safe connection and clipping behavior.	

The [documentation gaps](#) page provides more detail for handshake, storage, panel, MIDI, and wiring checks. After a result passes, update the relevant guide with the tested board/wiring revision and date; keep untested claims labeled accordingly.

Deployment smoke check — 2026-09-23

- USB identities: Raspberry Pi Pico W (RP2040, 2 MB flash; serial `E6614C311B886039`) and Arduino UNO R4 WiFi (serial `F412FA704F54`).
- Both sketches compiled and uploads verified. A full Pico flash backup was saved before upload; the new build uses the same 1 MB LittleFS region.
- Browser loaded the redesigned Perform/Shape/Explore/All controls panel at `http://quarkwave.local/`. It displayed **Pico: Connected** and **Uno: Connected · Pico patch** after startup, and selected the existing user patch **Test Pluck** in slot 3.
- Audio playback, control response, power-cycle recovery, standalone behavior, and MIDI Clock were **not tested** in this smoke check.

Owner-supplied hardware record — 2026-09-23

- The [board overview and close-up](#) show a Pico W, Uno R4 WiFi, bidirectional level-shifter module, and jumper wiring. The owner confirms Uno TX is shifted from 5 V to 3.3 V before Pico RX, with a common ground.
- **Power:** Each board is currently powered through its own USB port. The audio jack and `A0` output circuit have not been assembled.
- The [mono line-output schematic](#) is a proposal for a powered speaker or mixer line input. Its DC level, audio level, and sound are **not tested**. Record build and measurement results above once assembled.
- The owner also identified an unconnected [TDA1308 headphone amplifier module](#). Its pad ground, output DC, gain, and headphone sound have **not** been measured.

Automated browser and link check — 2026-09-23

- Live embedded page at `http://10.0.2.89/` matched the source HTML. Initial WebSocket state contained eight user slots, eight factory presets, user slot 3 "Test Pluck," 45 patch fields, and **Uno connected · Pico patch**.
- Headless Chrome exercised 50 control paths in their visible views, 41 pointer keys, and all 12 computer note keys at velocities 1, 100, and 127. The browser sent the expected WebSocket messages and Pico `patchData` reflected control changes. A held browser note survived view switches. These observations do **not** prove that the Uno received each MIDI message or produced the intended audio; there is no per-control Uno acknowledgment and the audio output is not built.
- The first run found that checked browser controls were parsed as zero by the Pico. The Pico handler now converts JSON booleans to 1/0; its sketch compiled, was uploaded, and a repeat control pass passed 50/50. The Pico source hash is recorded in the [index](#).
- Randomize returned all 45 patch fields. Factory and user loads selected the expected slots. Save As opened with eight destinations; factory save and occupied Save As without overwrite were rejected. Three visualization selections and Panic emitted their browser messages. Save/Save As success and Commit/Load Commit were **not run** to avoid altering the owner's stored patches. The existing user patch and live values were restored after the full sweep.

- An isolated explicit **Sync Pico patch to Uno** sometimes led to an Uno status timeout. A repeat sequence showed `pico → syncing → offline`; another earlier sequence later reported `syncFailed`. Restarting the Pico with the verified firmware led to repeated Wi-Fi connection attempts. A USB power cycle restored its web page, but the Uno still showed offline. A subsequent USB power cycle of the Uno restored **Uno connected · Pico patch**. The final browser snapshot showed user slot 3 “Test Pluck.” Treat patch sync and return MIDI reliability as **failed/unresolved** until diagnosed on hardware.

Uno LED display photograph — 2026-09-23

The owner supplied a [live photograph](#) of the Uno R4 WiFi matrix with several top-row pixels illuminated. The photograph confirms a powered, illuminated matrix, but one still frame cannot validate heartbeat timing, connection-indicator meaning, voice bars, VU/Scope response, or scrolling messages. Those behaviors are source-reviewed in the [LED display guide](#) and remain candidates for a timed hardware check.

Return-link parser fix — 2026-09-23

- **Reproduced:** one explicit sync caused two Uno reset messages, showing the Pico had retried the complete patch transfer. A later run showed Uno `Hello` reception every second while the Pico declared it offline.
- **Isolated:** a temporary Pico diagnostic build read valid raw Uno frames such as `F0 7D 00 11 02 F7`, but its generic MIDI receive library reported each as a two-byte SysEx. The Uno was sending status; the Pico discarded it before updating the connection state. The temporary diagnostic builds were removed after testing.
- **Fix deployed:** the Pico now directly validates the fixed six-byte Uno status and completion frames on its return UART. The Uno also drains up to 48 queued Pico UART bytes per loop so full patch transfers are serviced promptly. Both final sketches compiled and uploads verified. Their hashes are in the [index](#).
- **Retest:** the first isolated sync completed with one Uno reset and completion marker, and the Pico returned to **Connected · Pico patch** in about 0.2 seconds. Five subsequent explicit syncs all completed and stayed connected; four reported completion in roughly 0.2–0.35 seconds, and one status confirmation arrived after about 2.1 seconds without an Uno reset retry. After the Uno release build was installed, clean-start automatic patch sync and a final explicit sync passed.
- **Formerly failing action sequence:** headless Chrome ran Randomize, factory Load, user Load, explicit Sync, visualization commands, Panic, Save As dialog, and rejected save checks on the final builds. After ten seconds, status remained **connected / Pico patch**, with no sync failure. The original user slot 3 “Test Pluck” and all 45 live patch fields were restored; saved user files were not overwritten.
- **Still unverified:** exact shifter voltage, forced loss of an acknowledgment, unplugged return wire, audible sound, and per-control Uno sound response.

LED and audio timing diagnostic — 2026-09-23

- **Observed with the browser and Uno:** selecting Status, VU, or Scope reached the Uno (the `viz:` label scrolled). The Scope baseline appeared. Browser key presses flashed the Pico UART receive indicator in top-row column 7. The owner did not see the brief Status note flash in column 10. The Pico WebSocket reported **Uno connected · Pico patch** and the current patch had arpeggiator Off.

- **Temporary Uno timing build:** a three-second browser Note On reached the Uno and triggered one voice; its envelope and pre-DAC visualization level increased. The same build counted about **216 audio samples per second** while the source target is **22,050 samples per second**. Timing instrumentation attributed about 0.88 seconds of each wall-clock second to `rtpMIDI.read()` with no RTP session. The underlying Wi-FiS3 UDP polling makes modem calls for both RTP ports. These are measurements of the temporary diagnostic build, not calibrated release-build audio output.
- **Isolation:** skipping RTP reads in the temporary build raised measured output to roughly **15,000–16,000 samples per second**. Initializing the DAC once and writing subsequent samples directly raised it to roughly **18,000–19,000**. A timer-driven audio prototype with RTP active starved the main loop and was rejected. The synchronous `matrixScrollOnce()` also pauses the main loop during a view label. These tests identify multiple timing constraints; none is a verified release fix.
- **Restoration:** the unchanged release Uno sketch was recompiled and uploaded after the experiments. The Pico WebSocket again reported `connected: true, syncFailed: false, mode: pico`, with user slot 3 still selected. The output circuit remains unbuilt, so audible behavior is untested.
- **Next engineering check:** redesign audio scheduling and network polling together, then measure actual DAC sample intervals, RTP note/clock delivery, voice envelopes, all three matrix modes, and Pico patch sync under load. Keep the matrix response issue open until the full two-board and audio test passes. [Uno loop and audio output](#), [matrix scroll](#), [LED display guide](#).

Uno matrix response fix — 2026-09-23

- **Code changes:** corrected the state-variable filter recurrence and cached its coefficients, made ADSR advance from attack to decay and sustain, made the note indicator a wall-clock 400 ms flash, replaced live scrolling labels with stationary two-letter labels, initialized the DAC once, and moved chorus tap modulation to the control rate. Idle Wi-FiS3 RTP polling now runs every 250 ms; a connected RTP session is polled every 50 ms. A 2 ms cap prevents a long audio catch-up burst after blocked work. [Uno source](#).
- **Hardware diagnostic:** with the Pico-selected `Test Pluck` patch and no RTP peer, one held browser note produced a finite filter state, a pre-DAC visualization level around 0.09–0.12, and an envelope that settled at sustain 0.147 before release. Status, VU, and Scope commands reached the Uno during the held note. The diagnostic counted approximately 22,050 audio updates per wall-clock second in that one-note test. A separate four-note test still measured about 17,300 updates/s against the 22,050 target; this remains an audio performance limitation.
- **Deployed build:** the clean Uno release sketch compiled and uploaded after the diagnostic. The Pico again reported `connected: true, syncFailed: false, mode: pico`, with user slot 3 selected. A release-build key and all-three-view smoke sequence finished with the same healthy Pico–Uno link. The internal signals and message paths were checked. The owner then observed a held key lighting a Status voice bar and flashing column 10, with VU and Scope both responding. The unbuilt audio output remains untested.
- **RTP caveat:** polling a connected RTP session every 50 ms reduces idle load but may coalesce high-rate clock pulses. RTP note, clock, transport, and four-voice audio timing still need a dedicated two-board test before claiming those paths are repaired.

Nonblocking matrix scroll — 2026-09-23

- Restored full scrolling labels for Status, VU, Scope, and RTP connection. Each frame is drawn during the normal LED update slot; MIDI and audio continue during the animation.

- A temporary timing build showed **Viz: VU** moving from x=6 through x=-34 while a held browser note maintained a pre-DAC level around 0.09–0.11. It counted about 19,700 audio updates/s during the scroll, matching an instrumented build of the previous stationary-label firmware. The diagnostic itself changes the measured rate, so this comparison does not establish the release audio sample rate.
- The clean scrolling build compiled and was installed on the Uno. The unbuilt audio output remains untested. [Uno matrix code](#).

Four-voice audio stress and optimization — 2026-09-23

A temporary Uno build counted successful DAC updates over a five-second window after boot and Pico patch sync. It triggered four held notes (48, 55, 60, 64) directly in the Uno engine at velocity 100. The effects stress used chorus mix, bitcrush mix, tremolo depth, drive, fold, and delay mix at 1.0, bitcrush at 4 bits with 16-sample hold, delay feedback 0.8, and tempo-synced delay. The current Pico boot patch supplied other settings. The same counter and test sequence were used for the before-and-after comparison; diagnostic builds are not calibrated DAC interval measurements.

Build and load	Updates in 5 s	Effective update rate
Previous firmware · four voices × three unison · all stress effects	44,220	8,844/s
Optimized firmware · four voices × three unison · all stress effects	65,656	13,131/s
Optimized firmware · four voices × three unison · effects off	73,596	14,719/s
Optimized firmware · four voices × one oscillator · all stress effects	95,761	19,152/s

The full-load improvement is about 48%, but every measured load above remains below the 22,050-updates/s target. The largest remaining cost is twelve active oscillators; all-effects processing also consumes time. The firmware still requests 22,050 updates/s and drops overdue work after a long block. These figures do not claim an achieved 22.05 kHz output during the stress test.

The release change keeps the delay's 8-bit, 11,025 Hz ring so its maximum synced time and current sound resolution are retained. A fixed delay rate also prevents its RAM use from growing with a future oscillator-rate change. A separate 44.1 kHz compile now fits RAM, whereas the prior source failed to link; this is a compile-only check, and the measured CPU load does not support changing the deployed rate. The tap length, envelope increments, waveform morph segment, bitcrush settings, and tremolo step now avoid redundant per-sample calculations; ring wrap uses a comparison instead of division. Waveform blend and delay ring arithmetic were checked against the previous equations over 100,000 morph cases and ring endpoints. The clean 22.05 kHz Uno sketch compiled, was uploaded, and the Pico reported `connected: true, syncFailed: false, mode: pico`; a brief browser note test completed. The line-output circuit is unbuilt, so audible quality and DAC timing remain unverified. [Uno source](#).

Four-voice performance refinement — 2026-09-23

The same five-second direct-voice stress test described above was repeated with four notes, three unison oscillators per note, and all listed effects stressed. The sketch's requested rate stayed **22,050 updates/s** and its control rate stayed **1 kHz**. The numbers below count completed DAC writes in temporary instrumented builds; they do not establish evenly spaced sample intervals or audible quality.

Candidate	Updates in 5 s	Effective update rate	Decision
Previous release	65,656	13,131/s	Comparison baseline
Integer oscillator phases, refreshed at control rate	81,236	16,247/s	Retained
Integer waveform blend added	91,971	18,394/s	Retained
Standard o3 optimization added	98,197	19,639/s	Retained
Bounded audio rounding and LFO math added	99,883	19,977/s	Retained
RTP-MIDI compiled out for comparison	107,318	21,464/s	Measurement only; RTP remains enabled
Control rate reduced to 500 Hz	100,074	20,015/s	Rejected; negligible gain with lower timing resolution

The retained changes move oscillator phases with 32-bit integer increments, calculate pitch bend, detune, vibrato, and glide increments at the existing 1 kHz control rate, blend waveform table values as integers, use bounded rounding for audio samples, and replace small-exponent LFO calculations with accurate polynomial/table approximations. A 100,000-case comparison found the integer waveform blend within **0.000116** of the previous normalized waveform; sine-table interpolation was within **0.000082** of `sin()` in a 100,000-phase comparison, and the small-exponent polynomial's maximum error over the two-semitone vibrato range was below **0.000000004**. Glide now receives phase-increment updates every millisecond; its audible behavior remains to be checked.

A filter tangent lookup, drive-curve lookup, and one-step wavefold simplification showed no useful rate gain and were discarded. Two attempts to remove Wi-Fi UDP modem queries raised the benchmark but failed a real RTP invitation, so neither is in the release. The standard RTP path accepted control and data invitations, a test RTP note marked the Uno as **standalone sound** on the Pico, and an explicit patch sync returned to **Pico patch**. The clean optimized Uno sketch compiled and was uploaded. A browser note plus Status, VU, and Scope commands completed with the Pico reporting `connected: true, syncFailed: false, mode: pico`.

Open limit: the demanding four-voice/all-effects case still produced about **19,977 updates/s**, below the **22,050/s** target. With RTP compiled out it reached about **21,464/s**, also below target. Connected RTP-MIDI and actual DAC interval timing have not been characterized, and the audio output circuit is still unbuilt. Do not describe 22.05 kHz as achieved under all loads. Further work needs cycle-level profiling and a real-time audio scheduling/network design that preserves RTP clock and MIDI latency. [Uno source](#).

Pico RTP-MIDI gateway smoke check — 2026-09-23

Both current sketches compiled. The Uno upload completed; the Pico's serial upload entered its USB bootloader but did not expose a mounted drive to `arduino-cli`, so `picotool load -v -x` was used to verify and start the compiled Pico image while preserving the configured LittleFS partition. The Pico then served HTTP 200 from its recorded local IP. A local RTP-MIDI test sent a control invitation to UDP 5004, received AppleMIDI `OK`, sent the data invitation, and sent Note On/Off. A simultaneous browser WebSocket changed from `Uno: Connected · Pico patch` to `Uno: Connected · standalone sound`, and its new `rtpConnected` field changed from false to true. An explicit `syncUno` request then produced `syncing` followed by `Connected · Pico patch` with `syncFailed: false`. A second test sent Start, 25 Clock pulses at approximately 120 BPM, and Stop; the Uno remained `Connected · Pico patch`, as expected because clock alone does not mark standalone sound. Closing each RTP session returned `rtpConnected` to false. The last check reported `connected: true, syncFailed: false, rtpConnected: false, mode: pico`. This confirms the local network parser, gateway activity marker, return link, and explicit-sync acknowledgment; it does not establish audible note output or clock accuracy. The physical DIN connector is still unbuilt.

Uno network-free performance and wireless disconnect — 2026-09-23

The prior five-second direct-voice stress was repeated on the Uno with both `USE_WIFI_STACK=0` and `USE_RTP_MIDI=0`. Four notes (48, 55, 60, 64) used three unison oscillators each. Chorus mix, bitcrush mix, tremolo depth, drive, fold, and delay mix were 1.0; bitcrush used 4 bits and a 16-sample hold; delay feedback was 0.8 with tempo sync. The current Pico boot patch supplied other settings. The temporary build counted **110,251 DAC writes in 5,000 ms**, or **22,050.2 writes/s**. The extra write is a window-edge count; this meets the requested *average update count* of 22,050/s for this diagnostic.

A second temporary build instrumented scheduling lateness. It counted **110,250 writes in 5,000 ms**, with **6,775** writes starting over 45 μ s late, **2,381** over 100 μ s late, a **1,734 μ s** worst observed lateness, and **zero** audio catch-up resets. Instrumentation itself adds timing cost. A comparison with matrix redraws suppressed counted 110,248 writes, 5,032 over 45 μ s late, 2,214 over 100 μ s late, and a 1,723 μ s worst lateness. The small improvement did not justify degrading the LED display; normal redraws remain enabled. These counters show throughput and scheduling behavior in diagnostic builds, not DAC waveform timing or audible quality. The line-output circuit is still unbuilt.

The Pico and Uno were updated with source-aware note and sustain handling. The Pico AppleMIDI session now accepts one wireless controller, tracks held notes and pedal state by MIDI channel, and sends source-tagged releases on disconnect. The Uno tracks browser, DIN, and wireless ownership separately. A live local RTP test received an invitation and forwarded a note; a controller `BY` disconnect changed `rtpConnected` from true to false. A temporary Uno diagnostic logged one Note On and one engine Note Off for a wireless note released by disconnect. With a browser note held on the same pitch, it logged two Note Ons and only one engine Note Off after the later browser release, confirming that disconnect preserved browser ownership. A second diagnostic logged wireless pedal ownership changing `4 → 0` on disconnect and shared browser/wireless ownership changing `5 → 1 → 0` as the wireless session and then the browser pedal released. The production Uno image was then restored. Both updated sketches compiled and were uploaded; the release-build link was checked after restoration. A silent Wi-Fi loss without `BY` depends on the AppleMIDI session timeout and has not been timed on hardware. [Pico gateway](#), [Uno note state and audio loop](#).

Clearer Uno connection label — 2026-09-23

The browser's `mode: pico` status previously read **Uno: Connected · Pico patch**. The Pico/Uno protocol and mode value are unchanged; the panel now reads **Uno: Connected to Pico · sound loaded**. “Connected to Pico” describes the recent two-way MIDI handshake, and “sound loaded” describes the Uno's acknowledgment of the Pico's selected sound settings. The standalone, loading, awaiting-sound, offline, and sync-failure labels were adjusted to use the same language. The updated Pico sketch compiled and was uploaded. A live browser WebSocket reported `connected: true, mode: pico, syncFailed: false`, and the served HTML contained the new label. Fresh Perform, Shape, Explore, All controls, and staged Save As captures now show that wording. The captures do not establish audio output or save a patch. This is a historical label; the current source uses **Uno: Connected to Pico**. [Panel label, connection meaning](#).

Multi-bar Uno VU display — 2026-09-23

The previous VU view drew one horizontal output-level bar and a peak pixel. The new view draws **11 vertical bars** of recent mono pre-DAC output levels, oldest left and newest right, plus a peak dot in column 12. The top-row indicators and bottom-right timing diagnostic remain available. This is a short loudness history, not a frequency spectrum or a stereo meter. The LED guide's three-view SVG was refreshed as an illustration, not a photograph. [VU drawing](#), [LED guide](#).

The Uno sketch compiled. In the same temporary five-second four-voice, three-unison, all-effects stress described above, with VU selected, it completed **110,248 DAC writes in 5,000 ms** against the 110,250-write target. The normal firmware was restored and uploaded. A live browser `visualization` message selected mode `1`, the Uno USB log reported `[viz] viz: vu`, a browser test note was released, and the Pico still reported `connected: true, mode: pico, syncFailed: false`. This verifies the command and average update count; the moving bars have not yet been visually checked on the physical matrix, and the audio output circuit remains unbuilt.

Uno top-row status indicators — 2026-09-23

The Uno top row now uses columns 1–5 for Pico sound loaded, external MIDI used, valid external clock, sustain, and arpeggiator enabled. Column 8 briefly shows a voice steal, column 11 briefly shows an audio scheduler reset after more than 2 ms of lateness, and the note-start flash in column 10 is now drawn in Status, VU, and Scope. The existing Pico link, Pico receive, external activity, heartbeat, and bottom-right display timing signals remain. The top-row SVG and LED guide describe the exact conditions. [Uno source](#), [LED guide](#).

The Uno sketch compiled for **Arduino UNO R4 WiFi** (117,908 bytes flash, 16,300 bytes global RAM) and uploaded successfully over USB. The uploader reported the same serial port afterward. The local shell could not reach `quarkwave.local` or the previously recorded Pico IP, so the Pico handshake and the new physical LED patterns were not checked after this upload. Hardware acceptance still needs: sync a Pico patch; use an external MIDI controller; send a valid external clock; toggle sustain and arpeggiator; overlap five notes to see voice stealing; and exercise a demanding patch while watching for audio-slip flashes. A forced scheduler delay would test column 11, but that diagnostic should use a temporary build and the normal firmware should be restored afterward. Audible effects remain untested until the audio output is built.

Simpler Uno connection label — 2026-09-23

The browser source now displays **Uno: Connected to Pico** for its normal acknowledged-patch state; the previous label added “sound loaded.” The standalone, loading, awaiting-sound, offline, and failure labels remain distinct. The connection state and Pico–Uno MIDI protocol are unchanged. All five panel guide images were rendered again from the current embedded page with source-defined Warm Pad settings and a simulated connected status. The page was not uploaded in this change, so these images are not a live post-upload test. [Panel status](#), [guide index](#).

Logic Pro USB-audio owner check — 2026-09-23

After the corrected [USB-audio experiment](#) was running, the owner reported that it **worked beautifully** in Logic Pro using the Uno as the USB audio input. This confirms that the Mac/Logic route worked for this setup. The report did not specify the Logic project rate, buffer size, whether MIDI came from the Pico browser or an RTP session, or the duration and effects used. Do not treat those details as verified. The [Network MIDI setup guide](#) gives a repeatable Logic audio and optional RTP procedure for follow-up testing.

Ten-page onboard Help — 2026-09-23

The embedded panel now contains seven User Guide pages and three Technical details pages, all in the Pico's existing self-contained HTML. Browser checks on a source-rendered preview opened each page, verified the active navigation state and heading focus, held and released a computer-key note across page changes, and confirmed that a changed control value survived navigation. All ten topics fit desktop (1440 px), tablet (768 px), and phone (390 px) widths without document-level horizontal overflow. The final Pico sketch compiled with the expanded Help at 581,164 bytes of flash and 75,940 bytes of global RAM. Six guide images were rendered again with Warm Pad and simulated connection state. This Help update has **not** been uploaded to the Pico or accepted on physical hardware. The Markdown guides still hold the complete schematics, control-value tables, and test evidence; PDFs were not regenerated. [Embedded Help](#), [guide index](#).